Technology Architecture: UjenziXform TIS Integration with KRA eTIMS

Prepared for: Kenya Revenue Authority (KRA) — third-party integrator sandbox application **Platform:** UjenziXform (construction materials marketplace, Kenya) **TIS product name:** UjenziXform Trader Invoicing System (TIS) **Integrator service name:** UjenziXform KRA Invoicing Services **TIS product version:** 1.0.0 **Solution type:** OSCU (Online Sales Control Unit) — primary; VSCU initialization paths supported in code **Sandbox status: Not yet assigned** — UjenziXform is applying for KRA eTIMS sandbox access **Target sandbox URL:** `https://etims-api-sbx.kra.go.ke` **Certification status: Applicant / in development** — seeking KRA sandbox access and subsequent integrator certification **Document version:** 3.3 **Date:** 3 June 2026

> **One-page overview:** See `EXECUTIVE_SUMMARY.md` in this pack. **Sample API payload:** See `appendix/APPENDIX_SAMPLE_SALESREQ.md`.

> **Scope:** This document describes **how UjenziXform will integrate with KRA eTIMS** once KRA assigns sandbox access to us as a **potential third-party TIS integrator**. It is a forward-looking architecture submission — not a report of live sandbox operations.

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1. Purpose of this document

This document accompanies UjenziXform's **application for KRA eTIMS sandbox access** as a **potential third-party Trader Invoicing System (TIS) integrator**.

It explains:

- **What UjenziXform is** and why we require eTIMS integration
- **How we will connect to KRA eTIMS** after KRA issues sandbox credentials
- The **TIS software we have developed** in preparation (secure proxy, purchase-order mapping, fiscal receipt UI, audit, vendor onboarding)

We do **not** claim that KRA sandbox access or integrator certification has already been granted.

2. What UjenziXform is

UjenziXform is a web platform that connects:

User type	Role
Builders	Buy construction materials; create and accept purchase orders (POs)
Suppliers	Sell materials; hold catalog, pricing, and tax identity
Administrators	Operate the marketplace and TIS integrator tools

Typical transaction: a builder selects materials from a supplier, a **purchase order** is created and accepted, payment may follow (e.g. Paystack), and delivery workflows may apply. **KRA fiscal invoicing will be tied to the purchase order**, not to a separate internal invoice module.

The platform is designed so that the **eTIMS fiscal response** becomes the tax receipt for supplier sales once KRA sandbox testing and certification are complete.

3. What the TIS does

The **UjenziXform Trader Invoicing System (TIS)** will:

1. Collect **supplier tax identity** (KRA PIN, legal name, branch, device serial where required)
2. Map **purchase order line items** to KRA item codes and tax metadata
3. Submit **sales invoices** and **credit notes** to **KRA eTIMS** via REST (`POST /invoices`)
4. Store the **KRA/SCU response** (verification URL, control unit reference, JSON snapshot)
5. Display a **fiscal receipt with QR code** to buyers and suppliers
6. Log submissions for **admin review and retry**

TIS is the **eTIMS compliance layer** embedded in UjenziXform order workflows — not a standalone accounting product.

4. Integrator application status

Item	Value
Applicant name	UjenziXform KRA Invoicing Services
Platform	UjenziXform
Product name	UjenziXform Trader Invoicing System (TIS)
Product version	1.0.0
Solution type	OSCU
KRA eTIMS sandbox	**Not yet assigned — applying for access**
Target sandbox URL	`https://etims-api-sbx.kra.go.ke`
Target production URL	`https://etims-api.kra.go.ke` (only after KRA certification)
Integrator certification	**Not held** — applicant in development

UjenziXform is registering on the eTIMS portal and submitting this architecture as part of our request to be recognised as a **potential third-party TIS integrator** and to receive **KRA-assigned sandbox credentials**.

5. Parties and responsibilities

```
flowchart LR
    KRA[KRA eTIMS / SCU]
    INT[UjenziXform<br/>TIS integrator applicant]
    SUP[Supplier<br/>registered taxpayer]
    BUY[Builder<br/>buyer]

    INT -->|REST via etims-proxy<br/>KRA sandbox| KRA
    SUP -->|KRA PIN, catalog, branch, device| INT
    BUY -->|KRA PIN when B2B| INT
    SUP -->|Fiscal sale| BUY
```

Party	Responsibility
KRA	Sandbox access, eTIMS policy, SCU, integrator approval
UjenziXform	TIS software, secure API gateway, vendor onboarding, audit
Supplier	Accurate KRA PIN, legal name, item codes; issues fiscal sale
Builder	Optional KRA PIN; receives fiscal receipt for verification

Each invoice will carry the **supplier's KRA PIN** as `traderTin` — the **vendor taxpayer** is the issuer, not UjenziXform's PIN alone.

6. Integration architecture

Once KRA assigns sandbox access, integration will work as follows:

```
flowchart TB
    subgraph Users
        B[Builder]
        S[Supplier]
        A[Admin]
    end

    subgraph UjenziXform["UjenziXform platform"]
        UI[React web app]
        DB[(PostgreSQL)]
        LIB[src/lib/etims]
        EDGE[Edge Function etims-proxy]
    end

    subgraph KRA_side["KRA eTIMS sandbox"]
        SBX[etims-api-sbx.kra.go.ke]
        SCU[OSCU / SCU]
    end

    B & S & A --> UI
    UI --> DB
    UI --> LIB --> EDGE
    EDGE -->|HTTPS + KRA-issued credentials| SBX --> SCU
```

KRA-issued credentials will be stored as **server-side Edge secrets only**. The browser will never see them.

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7. Marketplace purchase flow → POST /invoices

Yes — POST /invoices is part of the current UjenziXform purchase flow. TIS does not receive orders from outside the platform. A **purchase order (PO)** is the business document; TIS maps that PO to KRA **SalesReq** and submits it via `POST /invoices`.

This section describes the **current marketplace design** (how buyers and suppliers work today). Once KRA assigns sandbox access, the same flow will call **KRA eTIMS sandbox** instead of any other endpoint.

7.1 End-to-end path (current design)

```
flowchart TB
    subgraph Buyer["Builder / buyer"]
        CART[Cart or Buy Now]
        PROFILE[Profile: KRA PIN, billing name]
    end

    subgraph Marketplace["UjenziXform marketplace"]
        PO[(purchase_orders)]
        GATE[Supplier tax identity check]
        ENRICH[Enrich lines with eTIMS item codes]
        BUILD[buildEtimsInvoiceBodyFromPurchaseOrder]
        PROXY[etims-proxy]
    end

    subgraph Supplier["Supplier taxpayer"]
        TAX[KRA PIN, legal name, catalog codes]
    end

    subgraph KRA["KRA eTIMS — upon sandbox assignment"]
        POST[POST /invoices]
    end

    CART --> GATE
    TAX --> GATE
    GATE -->|PO created confirmed or accepted| PO
    PROFILE --> BUILD
    PO --> ENRICH --> BUILD --> PROXY --> POST
    POST -->|verification URL, SCU data| PO
    PO --> RECEIPT[Fiscal receipt + QR to buyer and supplier]
```

7.2 Purchase paths on UjenziXform today

Path	What happens	Leads to POST /invoices?
Cart → Buy Now	Builder selects supplier, passes tax-identity check, PO created with status `confirmed`	**Yes** — automatic TIS submit after PO save (`CartSidebar`)
Materials grid → Buy Now	Direct single-item purchase, PO status `confirmed`	**Yes** — automatic TIS submit (`MaterialsGrid`)
Quote workflow	PO progresses through quote statuses until builder accepts	**When accepted** — PO must leave quote-pending statuses; supplier or admin can submit manually
Supplier Order Management	Supplier opens accepted PO	**Yes** — manual “Submit to KRA eTIMS” button
Admin TIS / eTIMS tools	Staff retry or test on a PO	**Yes** — manual submit / retry

Not wired today: automatic submit on quote accept (function exists in code but has no UI caller).

7.3 Gates before POST /invoices

Gate	Rule
Supplier tax identity	Valid `kra_pin` and legal business name (`assertSupplierTaxIdentityForCheckout`)
PO status	Blocked while `pending` , `draft` , or any `quote_*` / `quoted` status
Line item codes	Every PO line must resolve to an `etims_item_code` (from line JSON or catalog)
Buyer identity	`profiles.kra_pin` and billing name used as `customerPin` / `customerName` when present

If a gate fails, TIS **does not call** POST /invoices; the error is stored on `purchase_orders.etims_error` .

7.4 What POST /invoices sends (from the PO)

The PO is the single source for the integrator payload:

- **Header:** supplier KRA PIN (`traderTin`), supplier legal name, buyer PIN/name, PO number as `traderInvoiceNo` , totals, payment type, receipt type `S`
- **Lines:** `salesItems[]` from PO `items` JSON — quantity, unit price, amount, tax code, item code from catalog enrichment

Internal draft `invoices` rows on quote accept were **removed** (May 2026). The **eTIMS fiscal response** on the PO is the intended tax receipt for supplier sales.

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8. Sales invoice flow (KRA sandbox)

8.1 Steps (upon KRA sandbox assignment)

Step	Action
1	Supplier completes KRA PIN and legal business name
2	Catalog lines have KRA **item codes**
3	Builder's PO reaches **accepted/confirmed** status
4	TIS builds **SalesReq** from the PO
5	`etims-proxy` → `POST /invoices` on **KRA sandbox**
6	KRA returns SCU reference and **verification URL**
7	Response saved on PO; **fiscal receipt + QR** shown
8	Event logged in `tis_submission_log`

8.2 Sequence diagram

```
sequenceDiagram
    participant User as Builder / supplier / admin
    participant App as purchaseOrderEtims.ts
    participant DB as PostgreSQL
    participant Proxy as etims-proxy
    participant KRA as KRA eTIMS sandbox

    User->>App: Submit invoice for PO
    App->>DB: Load PO, supplier, buyer
    App->>App: Validate PO status, enrich item codes
    App->>App: buildEtimsInvoiceBodyFromPurchaseOrder()
    App->>DB: tis_submission_log (pending)
    App->>Proxy: POST invoices + SalesReq
    Proxy->>Proxy: JWT, role, allowlist
    Proxy->>KRA: POST /invoices (KRA credentials)
    alt Accepted
        KRA-->>Proxy: verification URL, SCU data
        App->>DB: etims_validated_at, etims_response
        App->>DB: tis_submission_log (success)
    else Rejected
        KRA-->>Proxy: error
        App->>DB: etims_error
        App->>DB: tis_submission_log (failed)
    end
```

8.3 Submission triggers (current design)

Trigger	TIS design
Order confirmation / checkout	Automatic submit after PO confirmed
Supplier manual submit	Supplier dashboard action
Admin operations	Admin TIS tools for retry and testing

PO statuses that **will block** submission until accepted: `pending`, `draft`, `quote_created`, `quote_received_by_supplier`, `quote_responded`, `quote_revised`, `quote_viewed_by_builder`, `quoted`.

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9. OSCU initialization flow

After KRA assigns sandbox access, vendor devices will be initialized via the admin **TIS Integrator Hub**:

```
sequenceDiagram
    participant Admin as Admin UI
    participant Init as tisOscuInitialization.ts
    participant Proxy as etims-proxy
    participant KRA as KRA eTIMS sandbox

    Admin->>Init: tin, bhfId, dvcSrlNo
    Init->>Proxy: POST selectInitOsdcInfo
    Proxy->>KRA: KRA sandbox credentials
    KRA-->>Init: cmcKey, dvcId, branch name
    Init-->>Admin: Success (key masked in logs)
```

Field	Meaning
`tin`	Vendor/supplier KRA PIN
`bhfId`	Branch code (e.g. `00`)
`dvcSrlNo`	KRA sandbox device serial

Communication keys will **never be stored in plain text** in the database.

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10. Vendor (supplier) onboarding flow

```
stateDiagram-v2
    [*] --> draft
    draft --> pending_review
    pending_review --> pending_kra
    pending_kra --> active
    pending_review --> rejected
    active --> suspended
    suspended --> active
```

Administrators will track each supplier in `tis_vendor_onboarding` . Status `pending_kra` covers KRA-related steps (device init, item registration) under UjenziXform's integrator application.

Before first sandbox invoice: valid supplier KRA PIN, legal name, item codes on all PO lines, buyer PIN when B2B.

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11. Credit notes

Credit notes will use the same `POST /invoices` endpoint on KRA sandbox with `receiptTypeCode: R` and `traderOrgInvoiceNo` referencing the original sale. Requires a prior successful sale on the PO.

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12. Master data and catalog

Before sandbox sales, each PO line will require a KRA **item code** (from PO line, `supplier_product_prices` , `materials` , or admin catalog). Item registration (`POST /items`), customer registration (`POST /customers`), and reference-data sync will be performed against **KRA sandbox** through the allowlisted proxy and admin Integrator API console.

13. Technical architecture layers

```
flowchart LR
    UI[Presentation layer] --> APP[Application layer<br/>purchaseOrderEtims, etc.]
    APP --> INT[Integration layer<br/>etims-proxy + allowlist]
    APP --> DB[(Persistence layer)]
    INT --> KRA[KRA eTIMS sandbox]
```

Layer	Technology
Frontend	React 18, TypeScript, Vite
Backend	Supabase PostgreSQL, Auth, Row Level Security
Gateway	Supabase Edge Function `etims-proxy`
Client library	`src/lib/etims/*`

14. API integration (etims-proxy)

All outbound KRA traffic will use one Edge Function calling `{ETIMS_BASE_URL}/{path}` with HTTP Basic authentication.

Planned configuration after sandbox assignment:

Edge secret	Value
`ETIMS_BASE_URL`	`https://etims-api-sbx.kra.go.ke` (no trailing slash)
`ETIMS_BASIC_USER`	Issued by KRA
`ETIMS_BASIC_PASSWORD`	Issued by KRA

Allowlisted paths match KRA OSCU REST specification: initialization, reference data, items, customers, invoices, stock/purchase/import sync (`supabase/functions/_shared/etimsPathAllowlist.ts`).

Access controls: JWT required; roles `supplier` , `admin` , or `super_admin` ; SSRF path allowlist; 400 KB body limit.

15. Sales invoice payload mapping

Integrator field	UjenziXform source
`traderInvoiceNo`	PO number
`traderTin` / `sellerPin`	Supplier `kra_pin`
`traderName`	Supplier legal / company name
`customerPin`	Buyer `profiles.kra_pin`
`salesItems[].itemCode`	Catalog / PO `etims_item_code`
`receiptTypeCode`	`S` (sale) or `R` (credit note)
`paymentType`	Supplier default or `01`
`salesDate`	`yyyyMMddHHmmss` at submission
`currency`	Default `KES`

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16. Data stored in the platform

Store	Purpose
`purchase_orders` (`etims_*`)	Fiscal submission outcome per order
`suppliers`	Vendor tax identity, branch, device serial
`profiles`	Buyer KRA PIN and billing identity
`tis_integrator_platform`	Integrator metadata and certification checklist
`tis_vendor_onboarding`	Per-supplier onboarding lifecycle
`tis_submission_log`	Submission audit trail

Not stored: KRA credentials, communication keys in plain text, secrets in browser code.

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17. Multi-vendor (multi-taxpayer) model

UjenziXform will operate **one integrator registration** and serve **many supplier taxpayers** on the same marketplace. Each invoice will use that supplier's KRA PIN as `traderTin`.

```
flowchart TB
    subgraph Integrator["UjenziXform integrator applicant"]
        HUB[TIS Integrator Hub]
        PROXY[etims-proxy]
    end

    subgraph Vendors["Onboarded suppliers"]
        V1[Supplier A - KRA PIN]
        V2[Supplier B - KRA PIN]
    end

    PO[Purchase order]
    V1 & V2 --> PO
    PO -->|traderTin = supplier PIN| INV[POST /invoices]
    PROXY --> INV
    INV --> KRA[KRA eTIMS sandbox]
```

18. Security controls

Control	Implementation
Credential isolation	KRA credentials in Supabase Edge secrets only
Authentication	Supabase JWT on every proxy call
Authorization	Roles: supplier, admin, super_admin
SSRF prevention	Strict path allowlist
RLS	TIS admin tables via `is_tis_integrator_staff()`
Key handling	OSCU communication keys redacted in audit logs
Request limit	400 KB max proxy body

19. Fiscal receipt shown to buyers

After a successful KRA submission, `EtimsFiscalReceiptView` will display:

- Issuer name and KRA PIN
- Trader invoice number and SCU-related fields
- Line items and totals
- **Verification URL** and **QR code** for KRA public validation

20. Audit, errors, and retry

Mechanism	Purpose
`tis_submission_log`	Per-submission audit (type, status, error, response)
`purchase_orders.etims_error`	Latest failure on the order
Admin Submission Ops	Staff retry and monitoring
Supplier dashboard	Manual submit and credit note

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21. Post-sandbox-assignment plan

Once KRA issues sandbox credentials, UjenziXform will:

Step	Action
1	Configure Edge secrets with KRA-issued sandbox credentials
2	Set `ETIMS_BASE_URL` to `https://etims-api-sbx.kra.go.ke`
3	Set platform environment to `sandbox_testing`
4	Complete OSCU initialization with KRA sandbox device serials
5	Register test items and customers on sandbox
6	Execute sales invoice and credit note test cases
7	Complete internal certification checklist
8	Submit sandbox test results to KRA for integrator approval
9	Production cutover only after formal KRA certification

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22. Implementation readiness matrix

Capability	Readiness	Notes
Secure `etims-proxy` gateway	**Ready**	Awaiting KRA sandbox URL and credentials
PO → sales invoice / credit note mapping	**Ready**	To be validated on KRA sandbox
Fiscal receipt + QR UI	**Ready**	
OSCU initialization flow	**Ready**	To run on KRA sandbox
Vendor onboarding workflow	**Ready**	
Submission audit logging	**Ready**	
Certification checklist tracking	**Ready**	Admin TIS Integrator Hub
KRA sandbox access	**Pending**	**Subject of this application**
KRA integrator certification	**Pending**	After successful sandbox testing
Production eTIMS	**Future**	After KRA approval

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23. KRA certification checklist mapping

To be completed on **KRA-assigned sandbox** after access is granted:

ID	Requirement
`platform_registered`	eTIMS portal integrator registration
`sandbox_access`	KRA-issued sandbox credentials
`oscu_vscu_init`	Device initialization on KRA sandbox
`master_data_sync`	Reference data endpoints
`item_registration`	POST /items
`customer_registration`	POST /customers
`sales_invoice`	POST /invoices type S
`credit_note`	POST /invoices type R
`fiscal_receipt_ui`	Receipt + QR display
`vendor_tis_identity`	Per-supplier PIN, branch, device
`vendor_onboarding_workflow`	Onboarding lifecycle
`submission_audit`	Logging and retry
`secure_credentials`	Server-side Edge secrets
`production_cutover`	After KRA certification

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24. Source code references

Component	Path
Edge proxy	`supabase/functions/etims-proxy/index.ts`
Path allowlist	`supabase/functions/_shared/etimsPathAllowlist.ts`
PO → invoice	`src/lib/etims/purchaseOrderEtims.ts`
OSCU init	`src/lib/etims/tisOscuInitialization.ts`
Audit log	`src/lib/etims/logTisSubmission.ts`
Admin hub	`src/components/admin/tis-integrator/TisIntegratorHub.tsx`
Platform DB	`supabase/migrations/20260525120000_tis_integrator_platform.sql`

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25. Appendix — sample SalesReq payload

KRA reviewers often require a concrete example of the **`POST /invoices`** body. UjenziXform provides a separate appendix (anonymised, illustrative PINs and item codes):

File	Contents
`appendix/APPENDIX_SAMPLE_SALESREQ.md`	HTTP call, field mapping, credit note variant, response shape
`appendix/APPENDIX_SAMPLE_SALESREQ.json`	Full sales invoice JSON (two PO lines)

Summary: A confirmed UjenziXform PO (e.g. `PO-2026-004821`) maps to SalesReq with supplier `traderTin`, buyer `customerPin`, and `salesItems[]` from catalog `etims_item_code` values. See appendix for the complete payload and PO-field mapping table.

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26. Document control

Version	Date	Changes
1.0	2026-06-03	Initial draft
2.0	2026-06-03	KRA presentation: business context, maturity matrix
3.0	2026-06-03	Sandbox application framing
3.1	2026-06-03	Application-only narrative
3.2	2026-06-03	Marketplace purchase flow → POST /invoices
3.3	2026-06-03	Executive summary + SalesReq appendix (separate files)

This document describes how UjenziXform will integrate with KRA eTIMS as a potential third-party TIS integrator after sandbox access is granted. For authoritative API contracts, refer to the official KRA eTIMS OSCU/VSCU Technical Specification.